

CodroidPython SDK Manual

Version: 2.1.7 | Package: `codroid-robot-sdk` | Python: 3.7+

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Requirements

Item	Requirement
Python	3.7+ (CPython / PyPy, 3.7 ~ 3.14)
OS	Linux, Windows, macOS
Runtime deps	None (optional <code>colorama</code> for colored terminal output)

Install

```
pip install codroid-robot-sdk
```

Optional colored terminal output:

```
pip install "codroid-robot-sdk[color]"
```

Verify Installation

```
python3 -c "import codroid; print(codroid.__version__)"
```

API Naming Convention

All public methods use **PascalCase** (consistent with C# / C++ SDKs).

```
# Correct
robot.ConnectRemoteAndSwitchOn()
di = robot.GetDi(0)

# Wrong – snake_case removed in 2.1.1
robot.connect_remote_and_switch_on() # does not exist
```

Unit Conventions

Layer	Linear	Angular
SDK public API	mm	deg (degrees)
TCP JSON protocol	mm	deg
CRI UDP binary (wire)	m	rad (radians)
<code>CriRealTimeData</code> (parsed)	mm	deg

`CriRealtimePacketParser.parse()` and `CriRealtimeDispatcher (convert_to_si=True)` automatically handle m ↔ mm and rad ↔ deg conversion.

Quick Start

Install

Via pip

```
pip install codroid-robot-sdk
```

From Source

```
git clone https://github.com/guybod/CodroidSDK.git
cd CodroidSDK/CodroidPython
pip install -e .
```

Minimal Example

Connect to the controller, enter remote mode, and enable.

```
from codroid import CodroidControlInterface, InitConsoleUtf8

InitConsoleUtf8() # Windows cmd UTF-8 fix; no-op on Linux

ROBOT_IP = "192.168.1.136"

with CodroidControlInterface(host=ROBOT_IP) as robot:
    robot.EnterRemoteModeViaAuto()
    robot.SwitchOn()
```

Run:

```
python3 demo.py
```

Complete Workflow Example

```
from codroid import (
    CodroidControlInterface,
    JointPoint,
    CartesianPoint,
    MoveInstruction,
    InitConsoleUtf8,
)

InitConsoleUtf8()

ROBOT_IP = "192.168.1.136"

with CodroidControlInterface(host=ROBOT_IP) as robot:
```

```
# 1. Connect and power on
robot.EnterRemoteModeViaAuto()
robot.SwitchOn()

# 2. IO operations
di0 = robot.GetDi(0)
robot.SetDo(10, di0)

# 3. Registers
reg_val = robot.GetRegisterValue(0)
robot.SetRegisterValue(0, reg_val + 1)

# 4. Joint motion
robot.MovJ(JointPoint.Degrees([0, 0, 90, 0, 90, 0]), speed=40, acceleration=100)

# 5. Linear motion
robot.MovL(CartesianPoint.MmDeg([400, 200, 500, 180, 0, 90]),
           speed=150, acceleration=500)
```

Using CodroidClient

`CodroidClient` inherits from `CodroidSession`, uses a background thread for receiving data, and supports publish/subscribe event dispatch.

```
from codroid import CodroidClient, InitConsoleUtf8

InitConsoleUtf8()

with CodroidClient(host="192.168.1.136") as robot:
    robot.EnterRemoteModeViaAuto()
    robot.SwitchOn()
```

Running Examples

```
# Basic usage
PYTHONPATH=src python examples/01_basic_usage.py --robot 192.168.8.136

# Motion examples
PYTHONPATH=src python examples/08_move.py --robot 192.168.8.136

# Blocking motion
PYTHONPATH=src python examples/15_sync_motion.py --robot 192.168.8.136

# Robot settings
PYTHONPATH=src python examples/14_robot_parameters.py --robot 192.168.8.136
```

Error Handling

All TCP commands throw exceptions on failure:

Exception	Condition
<code>CodroidError</code>	Base exception
<code>CodroidCommandException</code>	Controller returned <code>err</code> field
<code>CodroidNetworkError</code>	TCP connection or communication failure
<code>CodroidTimeoutError</code>	Operation timeout

```
from codroid import CodroidControlInterface, CodroidError, CodroidTimeoutError

try:
    with CodroidControlInterface(host="192.168.1.136") as robot:
        robot.EnterRemoteModeViaAuto()
        robot.SwitchOn()
        robot.MovJ([0, 0, 90, 0, 90, 0], speed=40, acceleration=100)
except CodroidTimeoutError:
    print("Operation timed out")
except CodroidError as e:
    print(f"SDK error: {e}")
```

Windows Console UTF-8

When running examples with Chinese characters in `cmd` (not Windows Terminal), call at entry:

```
from codroid import InitConsoleUtf8

InitConsoleUtf8()
```

All `examples/*.py` call this at the top of `if __name__ == "__main__"`. No-op on Linux / macOS.

Core Concepts

Client Lifecycle

Typical lifecycle for `CodroidSession` (alias `CodroidControlInterface`) and `CodroidClient`:

```
from codroid import CodroidClient

# Option 1: with statement (recommended)
with CodroidClient(host="192.168.1.136") as robot:
    robot.EnterRemoteModeViaAuto()
    robot.SwitchOn()
    # ... use API ...
# Disconnect() called automatically

# Option 2: manual management
robot = CodroidClient(host="192.168.1.136")
robot.Connect()
try:
    robot.EnterRemoteModeViaAuto()
    robot.SwitchOn()
    # ... use API ...
finally:
    robot.Disconnect()
```

CodroidSession vs CodroidClient

Feature	CodroidSession	CodroidClient
Transport	<code>JsonStreamClient</code> (sync blocking)	<code>TransportClient</code> (background thread)
Request/Response matching	Sync send-receive	Async ID matching
Publish/Subscribe	Not supported	Supported
Use case	Simple scripts, one-shot ops	Continuous receive, event-driven

TCP Command Model

SDK communicates with the controller via TCP JSON. Each command flow:

1. SDK assigns auto-incrementing `id`
2. Sends request: `{"id": N, "ty": "command/path", "db": {...}}`
3. Controller responds: `{"id": N, "ty": "...", "db": {...}, "err": ...}`
4. SDK matches request and response by `id`

```
# SDK handles id assignment and matching internally
response = robot._send_command("Robot/switchOn", "")
# response = CommonResponse(id=1, ty="Robot/switchOn", db=None, err=None)
```

CommonResponse

All TCP commands return `CommonResponse`:

```
@dataclass
class CommonResponse:
    id: Union[int, str]    # Request ID
    ty: str               # Response type
    db: Optional[Any]     # Response data
    err: Optional[Any]    # Error info (None means success)

    @property
    def is_success(self) -> bool:
        return self.err is None
```

Error Handling

SDK defines four exception types:

Exception	Trigger
<code>CodroidError</code>	Base exception; parameter validation failure, illegal operations
<code>CodroidCommandException</code>	Controller returned <code>err</code> field (protocol-level error)
<code>CodroidNetworkError</code>	TCP connection failure, communication interruption
<code>CodroidTimeoutError</code>	Operation timeout (connection, CRI wait, blocking motion)

```
from codroid import (
    CodroidError,
    CodroidCommandException,
    CodroidNetworkError,
    CodroidTimeoutError,
)

try:
    with CodroidClient(host="192.168.1.136") as robot:
        robot.ConnectRemoteAndSwitchOn()
except CodroidNetworkError:
    print("Cannot connect to controller")
except CodroidTimeoutError:
    print("Connection timed out")
except CodroidCommandException as e:
    print(f"Controller error: {e}")
except CodroidError as e:
```

```
print(f"SDK error: {e}")
```

Thread Safety

- `CriData` property returns the current cached CRI snapshot, readable from any thread.
 - TCP methods (`GetDi`, `MovJ`, etc.) can be called from any thread, but concurrent calls on the same client instance are not supported.
 - `CriRealtimeDispatcher`'s `SendCommand` / `SendTrajectory` are thread-safe (UDP is stateless).
-

Publish / Subscribe

`CodroidClient` supports subscribing to controller push event topics:

```
from codroid import CodroidClient, PublishTopics

def on_robot_status(notification):
    print(f"Received {notification.ty}: {notification.db}")

with CodroidClient(host="192.168.1.136") as robot:
    sub = robot.SubscribePublishTopic(PublishTopics.ROBOT_STATUS, on_robot_status)
    # ... run ...
    sub.dispose() # Unsubscribe
```

Available topics in `PublishTopics`:

- `PROJECT_STATE` — Project state change
- `VAR_UPDATE` — Variable update
- `ROBOT_STATUS` — Robot status
- `ROBOT_POSTURE` — Robot posture
- `ROBOT_COORDINATE` — Robot coordinate
- `LOG` — Log
- `ERROR` — Error

CodroidSession / CodroidClient API Reference

`CodroidSession` (alias `CodroidControlInterface`) is the main session class encapsulating all TCP commands. `CodroidClient` inherits it and replaces the transport layer, adding publish/subscribe capabilities.

Constructor

CodroidSession

```
CodroidSession(  
    host: str = "192.168.1.136",  
    port: int = 9001,  
    local_ip: str = "192.168.1.150",  
    udp_port: int = 10086,  
)
```

Parameter	Type	Default	Description
host	str	"192.168.1.136"	Controller IP address
port	int	9001	TCP port
local_ip	str	"192.168.1.150"	Local IP (for CRI UDP push)
udp_port	int	10086	Local UDP listening port

CodroidClient

```
CodroidClient(  
    host: str = "192.168.1.136",  
    port: int = 9001,  
    local_ip: str = "192.168.1.150",  
    udp_port: int = 10086,  
    timeout: float = 10.0,  
)
```

Additional parameter:

Parameter	Type	Default	Description
timeout	float	10.0	TCP request timeout (seconds)

Properties

CriData

```
@property
def CriData(self) -> Optional[CriRealTimeData]
```

Latest CRI real-time data snapshot. Requires calling `StartListenUdp()` or `StartCriDataPush()` first.

```
robot.StartListenUdp()
robot.WaitForCriData()
data = robot.CriData
print(f"Joint angles: {data.joint_position}")
print(f"TCP pose: {data.tcp_pose}")
print(f"Is moving: {data.status.is_moving}")
```

Connection Management

Connect

```
def Connect(self) -> CodroidSession
```

Establish TCP connection. Returns self for chaining. Called automatically with `with` statement.

Disconnect

```
def Disconnect(self) -> None
```

Disconnect TCP and stop CRI receive. Called automatically with `with` statement.

Convenience Connect

ConnectRemoteAndSwitchOn

```
def ConnectRemoteAndSwitchOn(self) -> CommonResponse
```

Combined operation: `EnterRemoteModeViaAuto()` → `SwitchOn()`. Must `Connect()` first or be inside `with` block.

```
with CodroidClient(host="192.168.1.136") as robot:
    robot.ConnectRemoteAndSwitchOn()
```

EnterRemoteModeViaAuto

```
def EnterRemoteModeViaAuto(self) -> CommonResponse
```

Calls `ToAuto()` then `ToRemote()`.

EnterManualModeViaAuto

```
def EnterManualModeViaAuto(self) -> CommonResponse
```

Calls `ToAuto()` then `ToManual()`.

Mode Switching

SwitchOn / SwitchOff

```
def SwitchOn(self) -> CommonResponse  
def SwitchOff(self) -> CommonResponse
```

Enable / disable the robot.

ToRemote / ToManual / ToAuto

```
def ToRemote(self) -> CommonResponse  
def ToManual(self) -> CommonResponse  
def ToAuto(self) -> CommonResponse
```

Switch to remote / manual / auto mode.

ToSimulation / ToActual

```
def ToSimulation(self) -> CommonResponse  
def ToActual(self) -> CommonResponse
```

Switch to simulation / actual mode.

StartDrag / StopDrag

```
def StartDrag(self) -> CommonResponse  
def StopDrag(self) -> CommonResponse
```

Enter / exit drag-and-teach mode. Only available in remote or manual mode.

ClearSystemError

```
def ClearSystemError(self) -> CommonResponse
```

Clear system errors.

Non-Blocking Motion Commands

MovJ

```
def MovJ(
    self,
    target: Union[JointPoint, CartesianPoint, MovePoint],
    speed: float,
    acceleration: float,
    blend: Optional[float] = None,
    coor: Optional[Sequence[float]] = None,
    tool: Optional[Sequence[float]] = None,
) -> CommonResponse
```

Joint interpolation motion. Target can be `JointPoint` (sends jp) or `CartesianPoint` (sends cp+rj).

Parameter	Type	Description
target	JointPoint / CartesianPoint	Motion target
speed	float	Speed
acceleration	float	Acceleration
blend	float	Blend radius, default 0 (exact arrival)
coor	Sequence[float]	User coordinate [x, y, z, a, b, c]
tool	Sequence[float]	Tool coordinate [x, y, z, a, b, c]

```
robot.MovJ(JointPoint.Degrees([0, 0, 90, 0, 90, 0]), speed=40, acceleration=100)
robot.MovJ(CartesianPoint.MmDeg([400, 200, 500, 180, 0, 90]), speed=40, acceleration=100)
```

MovL

```
def MovL(  
    self,  
    target: Union[CartesianPoint, JointPoint, MovePoint],  
    speed: float,  
    acceleration: float,  
    blend: Optional[float] = None,  
    coor: Optional[Sequence[float]] = None,  
    tool: Optional[Sequence[float]] = None,  
) -> CommonResponse
```

Linear interpolation motion. Target can be `CartesianPoint` or `JointPoint`.

MovC

```
def MovC(  
    self,  
    middle: CartesianPoint,  
    target: CartesianPoint,  
    speed: float,  
    acceleration: float,  
    blend: Optional[float] = None,  
    coor: Optional[Sequence[float]] = None,  
    tool: Optional[Sequence[float]] = None,  
) -> CommonResponse
```

Circular arc motion. Both middle and target are `CartesianPoint`.

MovCircle

```
def MovCircle(  
    self,  
    middle: CartesianPoint,  
    target: CartesianPoint,  
    circle_num: int,  
    speed: float,  
    acceleration: float,  
    blend: Optional[float] = None,  
    coor: Optional[Sequence[float]] = None,  
    tool: Optional[Sequence[float]] = None,  
) -> CommonResponse
```

Full circle motion. `circle_num` is the number of revolutions.

Move

```
def Move(  
    self,  
    path: Union[MotionPath, List[MoveInstruction], List[Dict[str, Any]]],  
) -> CommonResponse
```

Multi-segment path execution. `List[MoveInstruction]` is recommended.

```
path = [  
    MoveInstruction.MovJ(JointPoint.Degrees([0, 0, 90, 0, 90, 0]), speed=40, acc=100),  
    MoveInstruction.MovL(CartesianPoint.MmDeg([400, 200, 500, 180, 0, 90]), speed=150,  
acc=500),  
    MoveInstruction.MovL(JointPoint.Degrees([0, 0, 0, 0, 0, 0]), speed=150, acc=500),  
]  
robot.Move(path)
```

Blocking Motion Commands

`*Sync` methods send the motion command then automatically poll CRI data until the robot stably reaches the target. CRI data push must be started first.

MovJSync

```
def MovJSync(  
    self,  
    target: Union[JointPoint, CartesianPoint],  
    speed: float,  
    acceleration: float,  
    wait: Optional[MotionWaitOptions] = None,  
    blend: Optional[float] = None,  
    coor: Optional[Sequence[float]] = None,  
    tool: Optional[Sequence[float]] = None,  
) -> bool
```

Blocking joint motion. Returns `True` on reaching the target.

```
robot.StartListenUdp()  
robot.WaitForCriData()  
robot.MovJSync(JointPoint.Degrees([0, 0, 90, 0, 90, 0]), speed=40, acceleration=100)
```

MovLSync

```
def MovLSync(  
    self,  
    target: Union[CartesianPoint, JointPoint],  
    speed: float,  
    acceleration: float,  
    wait: Optional[MotionWaitOptions] = None,  
    blend: Optional[float] = None,  
    coor: Optional[Sequence[float]] = None,  
    tool: Optional[Sequence[float]] = None,  
) -> bool
```

Blocking linear motion.

MovCSync

```
def MovCSync(  
    self,  
    middle: CartesianPoint,  
    target: CartesianPoint,  
    speed: float,  
    acceleration: float,  
    wait: Optional[MotionWaitOptions] = None,  
    blend: Optional[float] = None,  
    coor: Optional[Sequence[float]] = None,  
    tool: Optional[Sequence[float]] = None,  
) -> bool
```

Blocking circular arc motion.

MovCircleSync

```
def MovCircleSync(  
    self,  
    middle: CartesianPoint,  
    target: CartesianPoint,  
    circle_num: int,  
    speed: float,  
    acceleration: float,  
    wait: Optional[MotionWaitOptions] = None,  
    blend: Optional[float] = None,  
    coor: Optional[Sequence[float]] = None,  
    tool: Optional[Sequence[float]] = None,  
) -> bool
```

Blocking full circle motion.

MoveSync

```
def MoveSync(  
    self,  
    path: Union[MotionPath, List[MoveInstruction], List[Dict[str, Any]]],  
    wait: Optional[MotionWaitOptions] = None,  
) -> bool
```

Blocking path execution. Waits for CRI to confirm the last segment reached the target.

MotionWaitOptions

Controls blocking motion wait behavior:

```
@dataclass
class MotionWaitOptions:
    timeout: float = 60.0 # Overall timeout (seconds)
    poll_interval: float = 0.05 # Poll interval (seconds)
    cri_stale_timeout: float = 0.5 # CRI data stale threshold (seconds)
    settled_samples: int = 3 # Consecutive settled sample count
    joint_tolerance_deg: float = 0.2 # Joint tolerance (degrees)
    cartesian_position_tolerance_mm: float = 1.0 # Cartesian position tolerance (mm)
    cartesian_orientation_tolerance_deg: float = 1.0 # Cartesian orientation tolerance
    (degrees)
```

```
from codroid import MotionWaitOptions

opts = MotionWaitOptions(timeout=30.0, joint_tolerance_deg=0.5)
robot.MovJSync(JointPoint.Degrees([0, 0, 90, 0, 90, 0]),
               speed=40, acceleration=100, wait=opts)
```

Motion Control

PauseRobotMotion / ResumeRobotMotion

```
def PauseRobotMotion(self) -> CommonResponse
def ResumeRobotMotion(self) -> CommonResponse
```

Pause / resume current motion.

StopRobotMove

```
def StopRobotMove(self) -> CommonResponse
```

Stop current motion.

MoveTo Commands

MoveTo

```
def MoveTo(
    self,
    move_type: MoveToType,
    target: Optional[MoveToTarget] = None,
) -> CommonResponse
```

Move to a preset position. `MoveToType` enum values:

Value	Description
<code>STOP (-1)</code>	Stop MoveTo

Value	Description
HOME (0)	Home position
SAFE (1)	Safe position
CANDLE (2)	Candle position
PACK (3)	Pack position
JOINT (4)	Joint planned to target
LINEAR (5)	Linear planned to target
RESUME (6)	Program resume point

```
from codroid import MoveToType

robot.MoveTo(MoveToType.HOME) # Go to Home
robot.MoveTo(MoveToType.SAFE) # Go to Safe
```

Must call `MoveToHeartbeat()` every 0.5s during MoveTo motion.

MoveToHeartbeat

```
def MoveToHeartbeat(self) -> CommonResponse
```

MoveTo heartbeat. Must be called every 0.5s during MoveTo motion.

StopMoveTo

```
def StopMoveTo(self) -> CommonResponse
```

Stop current MoveTo motion.

Jog Commands

StartJog

```
def StartJog(
    self,
    mode: JogMode,
    index: int,
    speed: float,
    coor_type: JogCoorType = JogCoorType.USER,
    coor_id: int = 1,
) -> CommonResponse
```

Start jogging.

Parameter	Type	Description
mode	JogMode	JOINT (1) joint jog / LINEAR (2) linear jog
index	int	Joint number (1-6) or linear axis (1-6 for xyzabc)
speed	float	Speed (-1.0 ~ 1.0)
coord_type	JogCoordType	USER (0) / TOOL (1)
coord_id	int	User coordinate ID

```
from codroid import JogMode

robot.StartJog(JogMode.JOINT, index=1, speed=0.5) # Joint 1 positive jog
```

StopJog

```
def StopJog(self) -> CommonResponse
```

Stop jogging.

JogHeartbeat

```
def JogHeartbeat(self) -> CommonResponse
```

Jog heartbeat. Must be called every 0.5s during jogging.

Speed Rate

SetManualMoveRate / SetAutoMoveRate

```
def SetManualMoveRate(self, rate: int) -> CommonResponse
def SetAutoMoveRate(self, rate: int) -> CommonResponse
```

Set manual / auto move rate. rate range 1~100.

CRI Data Push

StartListenUdp

```
def StartListenUdp(self) -> None
```

One-click CRI data receive setup: stop old push → start local UDP listener → notify controller to start push.

```
robot.StartListenUdp()  
robot.WaitForCriData()  
data = robot.CriData
```

WaitForCriData

```
def WaitForCriData(self, timeout: float = 5.0) -> CriRealTimeData
```

Wait for the first CRI data packet. Throws `CodroidTimeoutError` on timeout.

StartCriDataPush

```
def StartCriDataPush(  
    self,  
    ip: str,  
    port: int,  
    duration: int = 100,  
    high_precision: bool = True,  
    mask: int = 0xFFFF,  
) -> CommonResponse
```

Manually start CRI UDP push. `port` range 10000–65534. `duration` is push period (ms).

StopCriDataPush

```
def StopCriDataPush(self, ip: Optional[str] = None, port: Optional[int] = None) ->  
CommonResponse
```

Stop CRI push.

StartCriControl

```
def StartCriControl(  
    self,  
    filter_type: CriFilterType = CriFilterType.NONE,  
    duration: int = 1,  
    start_buffer: int = 3,  
) -> CommonResponse
```

Start CRI real-time control mode. `duration` is command interval (ms, 1–16, must divide 1000).

StopCriControl

```
def StopCriControl(self) -> CommonResponse
```

Stop CRI real-time control.

IO Operations

GetDi / GetDo / GetAi / GetAo

```
def GetDi(self, port: int) -> int      # Digital input, port 0~15, returns 0 or 1
def GetDo(self, port: int) -> int      # Digital output, port 0~15, returns 0 or 1
def GetAi(self, port: int) -> float    # Analog input, port 0~3
def GetAo(self, port: int) -> float    # Analog output, port 0~3
```

```
di0 = robot.GetDi(0)
do5 = robot.GetDo(5)
ai0 = robot.GetAi(0)
```

SetDo / SetAo

```
def SetDo(self, port: int, value: int) -> CommonResponse    # value: 0 or 1
def SetAo(self, port: int, value: float) -> CommonResponse
```

```
robot.SetDo(10, 1)
robot.SetAo(0, 3.14)
```

GetIoValues

```
def GetIoValues(self, io_requests: List[Dict[str, Any]]) -> CommonResponse
```

Batch read IO. `io_requests` format: [{"type": "DI", "port": 0}, ...].

Register Operations

GetRegisterValue / GetRegisterValues

```
def GetRegisterValue(self, address: int) -> Any
def GetRegisterValues(self, addresses: List[int]) -> CommonResponse
```

```
val = robot.GetRegisterValue(0)
vals = robot.GetRegisterValues([0, 1, 2])
```

SetRegisterValue

```
def SetRegisterValue(self, address: int, value: Any) -> CommonResponse
```

```
robot.SetRegisterValue(0, 42)
```

SetExtendArrayType / RemoveExtendArray

```
def SetExtendArrayType(self, index: int, data_type: ExtendArrayType) -> CommonResponse
def RemoveExtendArray(self, index: int) -> CommonResponse
```

Set / remove extended array data type. `index` range 0~999.

Robot Settings

GetRobotParameters

```
def GetRobotParameters(self) -> RobotParameters
```

Get settings interface parameter snapshot. Returns `RobotParameters` containing tool frame table, payload frame table, coordinate frame table, and default IDs.

```
params = robot.GetRobotParameters()
print(f"Default tool: {params.default_tool_id}")
print(f"Default payload: {params.default_payload_id}")
for frame in params.tool:
    print(f"  Tool[{frame.id}]: x={frame.x}, y={frame.y}, z={frame.z}")
```

SetToolFrame / SaveToolFrames

```
def SetToolFrame(self, frame_id: int, frame) -> CommonResponse
def SaveToolFrames(self, frames) -> CommonResponse
```

`SetToolFrame` modifies a single tool frame (read-modify-write, `frame_id` 1~15 only). `SaveToolFrames` replaces the full table (16 items, id=0 must be all zeros).

```
from codroid import RobotFrame

robot.SetToolFrame(1, RobotFrame(id=1, x=100, y=0, z=50, a=0, b=0, c=0))
```

SetPayloadFrame / SavePayloadFrames

```
def SetPayloadFrame(self, frame_id: int, frame) -> CommonResponse
def SavePayloadFrames(self, frames) -> CommonResponse
```

`SetPayloadFrame` modifies a single payload frame (`frame_id` 1~15 only). `SavePayloadFrames` replaces the full table.

```
from codroid import RobotPayloadFrame

robot.SetPayloadFrame(1, RobotPayloadFrame(id=1, m=2.5, mx=0, my=0, mz=50))
```

SetUserCoordinateFrame / SaveUserCoordinateFrames

```
def SetUserCoordinateFrame(self, frame_id: int, frame) -> CommonResponse
def SaveUserCoordinateFrames(self, frames) -> CommonResponse
```

Modify single / replace full user coordinate frame table.

SetDefaultToolId / SetDefaultPayloadId / SetDefaultUserCoordinateId

```
def SetDefaultToolId(self, tool_id: int) -> CommonResponse
def SetDefaultPayloadId(self, payload_id: int) -> CommonResponse
def SetDefaultUserCoordinateId(self, coordinate_id: int) -> CommonResponse
```

Set default tool / payload / user coordinate ID. Range 0~15.

SetCollisionSensitivity

```
def SetCollisionSensitivity(self, sensitivity: int) -> CommonResponse
```

Set collision detection sensitivity. Range 0~100. Firmware 2.3.2.10+ only.

SetPayload

```
def SetPayload(self, payload_id: int) -> CommonResponse
```

Set current payload ID. Firmware 2.3.2.10+ only.

Project & Script

RunScript

```
def RunScript(
    self,
    main_script: str,
    sub_threads: Optional[Dict[str, str]] = None,
    sub_programs: Optional[Dict[str, str]] = None,
    interrupts: Optional[Dict[str, str]] = None,
    vars: Optional[Dict[str, Any]] = None,
) -> CommonResponse
```

Run Lua script.

```
robot.RunScript('print("hello")', vars={"speed": 100})
```

Run / RunByIndex / RunStep

```
def Run(self, project_id: str) -> CommonResponse
def RunByIndex(self, index: int) -> CommonResponse
def RunStep(self, project_id: str) -> CommonResponse
```

Run / single-step a project.

PauseProject / ResumeProject / StopProject

```
def PauseProject(self) -> CommonResponse
def ResumeProject(self) -> CommonResponse
def StopProject(self) -> CommonResponse
```

Pause / resume / stop a project.

EnterRemoteScriptMode

```
def EnterRemoteScriptMode(self) -> CommonResponse
```

Enter remote script mode.

SetStartLine / ClearStartLine

```
def SetStartLine(self, line: int) -> CommonResponse
def ClearStartLine(self) -> CommonResponse
```

Set / clear start line.

RS485 Communication

Rs485Init

```
def Rs485Init(
    self,
    baudrate: Union[RS485BaudRate, int],
    stop_bit: RS485StopBits = RS485StopBits.ONE,
    parity: RS485Parity = RS485Parity.NONE,
) -> CommonResponse
```

Initialize end-effector RS485.

```
from codroid import RS485BaudRate

robot.Rs485Init(RS485BaudRate.B115200)
```

Rs485Flush

```
def Rs485Flush(self) -> CommonResponse
```

Flush RS485 read buffer.

Rs485Read

```
def Rs485Read(self, length: int, timeout: int = 3000) -> CommonResponse
```

Read RS485 data. `length` max 128 bytes, `timeout` max 3000ms.

Rs485Write

```
def Rs485Write(self, data: Union[List[int], bytes]) -> CommonResponse
```

Write RS485 data. Max 127 bytes.

Kinematics

AposToCpos

```
def AposToCpos(  
    self,  
    jp: Sequence[float],  
    coor: Optional[Sequence[float]] = None,  
    tool: Optional[Sequence[float]] = None,  
    ep: Sequence[float] = [],  
) -> CommonResponse
```

Forward kinematics (joint → Cartesian). `jp` is 6 joint angles (degrees).

CposToApos

```
def CposToApos(  
    self,  
    cp: Sequence[float],  
    rj: Optional[Sequence[float]] = None,  
    ep: Sequence[float] = [],  
) -> CommonResponse
```

Inverse kinematics (Cartesian → joint). `cp` is `[x, y, z, a, b, c]`, `rj` is reference joint angles (default `[20, 20, 20, 20, 20, 20]`).

CalculateRelativePose

```
def CalculateRelativePose(  
    self,  
    pos: Sequence[float],  
    offset: Sequence[float],  
    coord_type: CoordinateType = CoordinateType.TOOL,  
    pos_coord: Optional[Sequence[float]] = None,  
    coord: Optional[Sequence[float]] = None,  
) -> CommonResponse
```

Cartesian coordinate offset calculation.

Publish / Subscribe (CodroidClient only)

SubscribePublishTopic

```
def SubscribePublishTopic(  
    self,  
    topic_ty: str,  
    handler: Callable[[PublishNotification], None],  
    tc_milliseconds: int = 100,  
) -> PublishTopicSubscription
```

Subscribe to controller push topic. Returns `PublishTopicSubscription`, call `.dispose()` to unsubscribe.

```
from codroid import CodroidClient, PublishTopics  
  
def on_status(notification):  
    print(f"{notification.ty}: {notification.db}")  
  
with CodroidClient(host="192.168.1.136") as robot:  
    sub = robot.SubscribePublishTopic(PublishTopics.ROBOT_STATUS, on_status)  
    # ... run ...  
    sub.dispose()
```

Debug

```
robot.debug = True # Print sent/received raw JSON
```

Motion API Reference

Motion Target Types

JointPoint

```
@dataclass
class JointPoint:
    jp: List[float] # 6 joint angles (degrees)
```

Joint target. Used at the business layer to declare "this is a 6-axis joint angle", avoiding confusion with TCP pose.

Factory Method

```
JointPoint.Degrees(joints_deg: Sequence[float]) -> JointPoint
```

Construct from six joint angles (degrees).

```
jp = JointPoint.Degrees([0, 0, 90, 0, 90, 0])
```

CartesianPoint

```
@dataclass
class CartesianPoint:
    cp: List[float] # TCP pose [x,y,z,rx,ry,rz] (mm + degrees)
    rj: Optional[List[float]] = None # IK reference joint angles (degrees)
```

Cartesian target. `cp` is required; `rj` is optional (defaults to controller default reference joints when packing).

Factory Methods

```
CartesianPoint.MmDeg(pose_mm_deg: Sequence[float]) -> CartesianPoint
```

TCP pose `[x,y,z,rx,ry,rz]` (mm + degrees), no reference joints.

```
CartesianPoint.MmDegWithRef(
    pose_mm_deg: Sequence[float],
    ref_joints_deg: Sequence[float],
) -> CartesianPoint
```

TCP pose + IK reference joints (degrees). Recommended when using `movJ/movL` to TCP and caring about orientation solution.

```

cp = CartesianPoint.MmDeg([400, 200, 500, 180, 0, 90])
cp_with_ref = CartesianPoint.MmDegWithRef(
    [400, 200, 500, 180, 0, 90],
    [0, 0, 90, 0, 90, 0],
)

```

MovePoint

```

@dataclass
class MovePoint:
    jp: Optional[Sequence[float]] = None
    cp: Optional[Sequence[float]] = None
    rj: Optional[Sequence[float]] = None
    ep: Optional[Sequence[float]] = None

```

General motion point definition. Usually not used directly; constructed via `JointPoint` / `CartesianPoint`.

Factory Methods

```

MovePoint.FromJoint(joint: JointPoint) -> MovePoint
MovePoint.FromCartesian(cart: CartesianPoint) -> MovePoint

```

Motion Instructions

MoveInstruction

```

@dataclass
class MoveInstruction:
    motion_type: MotionType
    target: MovePoint
    speed: float
    acc: float
    blend: Optional[float] = None
    relative_blend: Optional[float] = None
    middle: Optional[MovePoint] = None
    circle_num: Optional[int] = None
    coor: Optional[Sequence[float]] = None
    tool: Optional[Sequence[float]] = None

```

Single-segment `Robot/move` instruction. Use class methods to construct; do not manually write `type` + bare `MovePoint`.

Factory Methods

MoveInstruction.MovJ

```
MoveInstruction.MovJ(  
    target: Union[JointPoint, CartesianPoint],  
    speed: float,  
    acc: float,  
    blend: Optional[float] = None,  
    relative_blend: Optional[float] = None,  
    coor: Optional[Sequence[float]] = None,  
    tool: Optional[Sequence[float]] = None,  
) -> MoveInstruction
```

Joint motion movJ. Target can be joint or TCP.

MoveInstruction.MovL

```
MoveInstruction.MovL(  
    target: Union[CartesianPoint, JointPoint],  
    speed: float,  
    acc: float,  
    blend: Optional[float] = None,  
    relative_blend: Optional[float] = None,  
    coor: Optional[Sequence[float]] = None,  
    tool: Optional[Sequence[float]] = None,  
) -> MoveInstruction
```

Linear motion movL. Target can be TCP or joint.

MoveInstruction.MovC

```
MoveInstruction.MovC(  
    middle: CartesianPoint,  
    target: CartesianPoint,  
    speed: float,  
    acc: float,  
    blend: Optional[float] = None,  
    relative_blend: Optional[float] = None,  
    coor: Optional[Sequence[float]] = None,  
    tool: Optional[Sequence[float]] = None,  
) -> MoveInstruction
```

Circular arc motion movC. Both middle and target are TCP.

MoveInstruction.MovCircle

```
MoveInstruction.MovCircle(  
    middle: CartesianPoint,  
    target: CartesianPoint,  
    circle_num: int,  
    speed: float,  
    acc: float,  
    blend: Optional[float] = None,  
    relative_blend: Optional[float] = None,  
    coor: Optional[Sequence[float]] = None,  
    tool: Optional[Sequence[float]] = None,  
) -> MoveInstruction
```

Full circle motion movCircle.

MotionType

```
class MotionType(str, Enum):  
    MOVJ = "movJ"  
    MOVL = "movL"  
    MOVC = "movC"  
    MOVCIRCLE = "movCircle"
```

MoveTo Targets

MoveToTarget

```
@dataclass  
class MoveToTarget:  
    cp: Optional[Sequence[float]] = None  
    jp: Optional[Sequence[float]] = None  
    ep: Sequence[float] = field(default_factory=list)
```

MoveTo-specific target structure.

Factory Methods

```
MoveToTarget.Joint(joint: JointPoint) -> MoveToTarget  
MoveToTarget.Cartesian(cart: CartesianPoint) -> MoveToTarget
```

MoveToType

```
class MoveToType(IntEnum):
    STOP = -1      # Stop MoveTo
    HOME = 0       # Home position
    SAFE = 1       # Safe position
    CANDLE = 2     # Candle position
    PACK = 3       # Pack position
    JOINT = 4      # Joint planned
    LINEAR = 5     # Linear planned
    RESUME = 6     # Program resume point
```

Jog Parameters

JogMode

```
class JogMode(IntEnum):
    JOINT = 1      # Joint jog
    LINEAR = 2     # Linear jog
```

JogCoorType

```
class JogCoorType(IntEnum):
    USER = 0      # User coordinate
    TOOL = 1       # Tool coordinate
```

MotionPath (Legacy API)

```
class MotionPath:
    def add(self, instruction: MoveInstruction) -> MotionPath
    def MovJ(self, target, speed, acc, blend=None) -> MotionPath
    def MovL(self, target, speed, acc, blend=None) -> MotionPath
    def MovC(self, target, middle, speed, acc, blend=None) -> MotionPath
    def clear(self) -> None
    def get_commands(self) -> List[Dict[str, Any]]
```

Motion path builder. New code should prefer `List[MoveInstruction] + Move()`.

Complete Multi-Segment Path Example

```
from codroid import (
    CodroidClient,
    JointPoint,
    CartesianPoint,
    MoveInstruction,
    MotionWaitOptions,
```

```

    InitConsoleUtf8,
)

InitConsoleUtf8()

with CodroidClient(host="192.168.8.136") as robot:
    robot.ConnectRemoteAndSwitchOn()

    # Four-combination path
    path = [
        MoveInstruction.MovJ(JointPoint.Degrees([0, 0, 90, 0, 90, 0]),
                              speed=40, acc=100),
        MoveInstruction.MovJ(CartesianPoint.MmDeg([400, 200, 500, 180, 0, 90]),
                              speed=40, acc=100),
        MoveInstruction.MovL(CartesianPoint.MmDeg([400, -200, 500, 180, 0, 90]),
                              speed=150, acc=500),
        MoveInstruction.MovL(JointPoint.Degrees([0, 0, 0, 0, 0, 0]),
                              speed=150, acc=500),
    ]
    robot.Move(path)

    # Blocking path execution
    robot.StartListenUdp()
    robot.WaitForCriData()
    robot.MoveSync(path, wait=MotionWaitOptions(timeout=120.0))

```

Motion Parameter Notes

speed / acceleration

- `speed`: Speed value. Exact unit depends on motion type and controller configuration.
- `acceleration`: Acceleration value.

blend

Blend radius. Default `0.0` (exact arrival at target). When set to a value > 0 , the robot smoothly transitions near the target without stopping.

coord / tool

Optional user coordinate and tool coordinate, format `[x, y, z, a, b, c]` (mm + degrees).

Data Types & Enums

Communication Models

CommonResponse

```
@dataclass
class CommonResponse:
    id: Union[int, str]
    ty: str
    db: Optional[Any] = None
    err: Optional[Any] = None

    @property
    def is_success(self) -> bool:
        return self.err is None
```

TCP JSON common response. All TCP commands return this type.

Field	Type	Description
id	int / str	Request ID
ty	str	Response type
db	Any	Response data
err	Any	Error info (None means success)

Legacy alias: `CodroidResponse = CommonResponse`

CodroidRequest

```
@dataclass
class CodroidRequest:
    id: Union[int, str]
    ty: str
    db: Optional[Any] = None
```

SDK internal request structure.

CRI Real-time Data

CriRealTimeData

```
@dataclass
class CriRealTimeData:
    timestamp: int = 0
    status: CriStatus = field(default_factory=CriStatus)
    joint_pos: List[float] = field(default_factory=list)
    joint_vel: List[float] = field(default_factory=list)
    cartesian_pos: List[float] = field(default_factory=list)
    cartesian_vel: List[float] = field(default_factory=list)
    tcp_speed: float = 0.0
    joint_torque: List[float] = field(default_factory=list)
    external_torque: List[float] = field(default_factory=list)
    extra_axis_pos: List[float] = field(default_factory=list)
```

CRI parsed real-time data. All values converted to mm + degrees.

Field	Property Alias	Type	Unit	Description
timestamp	timestamp_ms	int	ms	Timestamp
status	—	CriStatus	—	Status bits
joint_pos	joint_position	List[float]	deg	6 joint angles
joint_vel	joint_velocity	List[float]	deg/s	6 joint velocities
cartesian_pos	tcp_pose	List[float]	mm, deg	TCP pose [x,y,z,rx,ry,rz]
cartesian_vel	tcp_velocity	List[float]	mm/s, deg/s	TCP velocity
tcp_speed	tcp_linear_velocity	float	mm/s	TCP linear speed
joint_torque	joint_output_torque	List[float]	Nm	Joint output torque
external_torque	joint_external_force	List[float]	Nm	External torque
extra_axis_pos	external_axis_position	List[float]	—	External axis position

Legacy alias: CRIData = CriRealTimeData

CriStatus

```
@dataclass
class CriStatus:
    status1_raw: int = 0
    status2_raw: int = 0
    project_running: bool = False
    project_stopped: bool = False
    project_paused: bool = False
    is_enabling: bool = False
```

```

is_disabled: bool = False
is_manual: bool = False
is_dragging: bool = False
is_moving: bool = False
collision_stop: bool = False
is_at_safe_pos: bool = False
has_alarm: bool = False
is_simulation: bool = False
is_emergency_stop: bool = False
is_rescue: bool = False
is_auto: bool = False
is_remote: bool = False
rt_control_mode: bool = False
error_code: int = 0

```

CRI status bit parsing.

Field	Description
project_running	Project running
project_stopped	Project stopped
project_paused	Project paused
is_enabling	Enabled
is_disabled	Disabled
is_manual	Manual mode
is_dragging	Drag mode
is_moving	Moving
collision_stop	Collision stopped
is_at_safe_pos	At safe position
has_alarm	Has alarm
is_simulation	Simulation mode
is_emergency_stop	Emergency stop pressed
is_rescue	Rescue mode
is_auto	Auto mode
is_remote	Remote mode
rt_control_mode	Real-time control mode
error_code	Error code (high 8 bits)

Legacy alias: `CRIStatus = CriStatus`

Robot Settings

RobotFrame

```
@dataclass
class RobotFrame:
    id: int
    x: float = 0.0
    y: float = 0.0
    z: float = 0.0
    a: float = 0.0
    b: float = 0.0
    c: float = 0.0
```

Tool / user coordinate single frame.

Field	Type	Description
id	int	Slot ID (0~15, 0 is reserved)
x, y, z	float	Position (mm)
a, b, c	float	Orientation (degrees)

```
frame = RobotFrame(id=1, x=100, y=0, z=50, a=0, b=0, c=0)
```

RobotPayloadFrame

```
@dataclass
class RobotPayloadFrame:
    id: int
    m: float = 0.0
    mx: float = 0.0
    my: float = 0.0
    mz: float = 0.0
```

Payload coordinate single frame.

Field	Type	Description
id	int	Slot ID (0~15, 0 is reserved)
m	float	Mass (kg)
mx, my, mz	float	Center of mass (mm)

```
payload = RobotPayloadFrame(id=1, m=2.5, mx=0, my=0, mz=50)
```

RobotParameters

```
@dataclass
class RobotParameters:
    default_tool_id: int = 0
    default_payload_id: int = 0
    default_coordinate_id: int = 0
    max_payload: float = 0.0
    tool: List[RobotFrame] = field(default_factory=list)
    payload: List[RobotPayloadFrame] = field(default_factory=list)
    coordinate: List[RobotFrame] = field(default_factory=list)
```

Settings interface parameter snapshot returned by `GetRobotParameters()`.

Field	Type	Description
default_tool_id	int	Default tool frame ID
default_payload_id	int	Default payload ID
default_coordinate_id	int	Default user coordinate ID
max_payload	float	Max payload (kg)
tool	List[RobotFrame]	Tool frame table (16 items)
payload	List[RobotPayloadFrame]	Payload frame table (16 items)
coordinate	List[RobotFrame]	User coordinate frame table (16 items)

Global Variables

GlobalVariable

```
@dataclass
class GlobalVariable:
    value: Any
    note: Optional[str] = None
```

Global variable data model.

Field	Type	Description
value	Any	Variable value (supports int, float, str, list, dict)
note	str	Variable note

```
var = GlobalVariable(value=42, note="counter")
robot.SaveGlobalVar("counter", var)
```

Enum Types

CoordinateType

```
class CoordinateType(str, Enum):
    USER = "user"
    TOOL = "tool"
```

Coordinate system type.

IOType

```
class IOType(str, Enum):
    DI = "DI" # Digital Input
    DO = "DO" # Digital Output
    AI = "AI" # Analog Input
    AO = "AO" # Analog Output
```

ExtendArrayType

```
class ExtendArrayType(str, Enum):
    BOOL = "Bool"
    UINT8 = "UInt8"
    INT8 = "Int8"
    UINT16 = "UInt16"
    INT16 = "Int16"
    UINT32 = "UInt32"
    INT32 = "Int32"
    FLOAT32 = "Float32"
```

Extended array data type.

RS485BaudRate

```
class RS485BaudRate(IntEnum):
    B110 = 110
    B300 = 300
    B600 = 600
    B1200 = 1200
    B2400 = 2400
    B4800 = 4800
    B9600 = 9600
    B14400 = 14400
    B19200 = 19200
    B38400 = 38400
    B56000 = 56000
```

```
B57600 = 57600
B115200 = 115200
B128000 = 128000
B230400 = 230400
```

RS485StopBits

```
class RS485StopBits(IntEnum):
    ONE = 1
    TWO = 2
```

RS485Parity

```
class RS485Parity(IntEnum):
    NONE = 0
    ODD = 1
    EVEN = 2
```

CriMask

```
class CriMask(IntFlag):
    TIMESTAMP = 1 << 0
    STATUS_1 = 1 << 1
    STATUS_2 = 1 << 2
    JOINT_POS = 1 << 8
    JOINT_VEL = 1 << 9
    CARTESIAN_POS = 1 << 10
    CARTESIAN_VEL = 1 << 11
    TCP_SPEED = 1 << 12
    JOINT_TORQUE = 1 << 13
    EXTERNAL_TORQUE = 1 << 14
    EXTRA_AXIS_POS = 1 << 15
```

CRI push bitmask.

Legacy alias: `CRIMask = CriMask`

CriFilterType

```
class CriFilterType(IntEnum):
    NONE = 0
    AVERAGE = 1
    LOW_PASS = 2
    ELLIPTIC = 3
```

CRI real-time control filter type.

Legacy alias: `CRIFilterType = CriFilterType`

Exceptions

CodroidError

```
class CodroidError(Exception):  
    pass
```

Base exception. Parameter validation failure, illegal operations, etc.

CodroidCommandException

```
class CodroidCommandException(CodroidError):  
    pass
```

Controller returned `err` field.

CodroidNetworkError

```
class CodroidNetworkError(CodroidError):  
    pass
```

TCP connection or communication failure.

CodroidTimeoutError

```
class CodroidTimeoutError(CodroidError):  
    pass
```

Operation timeout.

CRI Real-time Data & Control API Reference

CriRealtimePacketParser

```
class CriRealtimePacketParser:
    @staticmethod
    def parse(data: bytes) -> Optional[CriRealTimeData]
```

Parse 308-byte CRI UDP packets into `CriRealTimeData` (mm + degrees). Returns `None` for non-308-byte packets.

Automatic wire-layer unit conversion:

- Joint angles: rad → deg
- TCP position: m → mm
- TCP orientation: rad → deg
- Velocity: m/s → mm/s, rad/s → deg/s

```
from codroid import CriRealtimePacketParser

data = CriRealtimePacketParser.parse(raw_bytes)
if data is not None:
    print(f"Joint angles: {data.joint_position}")
    print(f"TCP pose: {data.tcp_pose}")
    print(f"Is moving: {data.status.is_moving}")
```

CriStreamHandler

```
class CriStreamHandler:
    def __init__(
        self,
        high_precision: bool = True,
        mask: int = 0xFFFF,
        joint_count: int = 6,
        extra_axis_count: int = 0,
    )
```

Variable mask/precision CRI UDP packet parser. Supports flexible bitmask and precision configuration.

Parameter	Type	Default	Description
<code>high_precision</code>	<code>bool</code>	<code>True</code>	Double (Float64) / Single (Float32) precision
<code>mask</code>	<code>int</code>	<code>0xFFFF</code>	Bitmask controlling which fields to parse
<code>joint_count</code>	<code>int</code>	<code>6</code>	Number of joints
<code>extra_axis_count</code>	<code>int</code>	<code>0</code>	Number of extra axes

bind

```
def bind(self, port: int) -> None
```

Bind UDP port.

parse_packet

```
def parse_packet(self, data: bytes) -> CriRealTimeData
```

Parse a single CRI data packet.

```
from codroid import CriStreamHandler

handler = CriStreamHandler(high_precision=True, mask=0xFFFF, joint_count=6)
handler.bind(10086)

try:
    while True:
        data, addr = handler._sock.recvfrom(2048)
        parsed = handler.parse_packet(data)
        print(f"Joint angles: {parsed.joint_position}")
finally:
    handler._sock.close()
```

CriRealtimeDispatcher

```
class CriRealtimeDispatcher:
    def __init__(
        self,
        controller_ip: str,
        controller_udp_port: int = 9030,
        convert_to_si: bool = True,
    )
```

UDP real-time command dispatcher. Sends 64-byte control commands to the controller.

Parameter	Type	Default	Description
<code>controller_ip</code>	<code>str</code>	—	Controller IP
<code>controller_udp_port</code>	<code>int</code>	9030	Controller UDP port
<code>convert_to_si</code>	<code>bool</code>	True	Auto-convert mm/deg to m/rad

Supports `with` statement.

SendCommand

```
def SendCommand(self, position6: Sequence[float], space: TrajectorySpace) -> None
```

Send a single real-time control frame.

Parameter	Type	Description
position6	Sequence[float]	6 position values (mm+deg or deg)
space	TrajectorySpace	JOINT or CARTESIAN

SendTrajectory

```
def SendTrajectory(  
    self,  
    trajectory: Iterable[TrajectoryPoint],  
    space: TrajectorySpace,  
    period_ms: int,  
) -> None
```

Send a complete trajectory. Sends frame by frame at `period_ms` intervals.

```
from codroid import CriRealtimeDispatcher, TrajectorySpace  
  
with CriRealtimeDispatcher("192.168.1.136") as dispatcher:  
    dispatcher.SendCommand([0, 0, 90, 0, 90, 0], TrajectorySpace.JOINT)
```

TrajectoryGenerator

```
class TrajectoryGenerator:  
    @staticmethod  
    def generate(  
        start: Sequence[float],  
        target: Sequence[float],  
        request: TrajectoryRequest,  
    ) -> List[TrajectoryPoint]  
  
    @staticmethod  
    def generate_multi_segment(  
        waypoints: Sequence[Sequence[float]],  
        request: TrajectoryRequest,  
    ) -> List[TrajectoryPoint]
```

Offline trajectory generation.

generate

Single-segment trajectory generation. Both `start` and `target` are 6-element arrays.

generate_multi_segment

Multi-segment trajectory generation. `waypoints` must contain at least 2 waypoints.

```
from codroid import (
    TrajectoryGenerator,
    TrajectoryRequest,
    TrajectorySpace,
    TrajectoryProfile,
)

request = TrajectoryRequest(
    space=TrajectorySpace.JOINT,
    frequency_hz=100,
    profile=TrajectoryProfile.CUBIC,
    acceleration=100,
    speed=40,
)

trajectory = TrajectoryGenerator.generate(
    [0, 0, 0, 0, 0, 0],
    [0, 0, 90, 0, 90, 0],
    request,
)

for point in trajectory:
    print(f"t={point.t:.3f}s, pos={point.position}")
```

TrajectoryRequest

```
@dataclass
class TrajectoryRequest:
    space: TrajectorySpace
    frequency_hz: float
    profile: TrajectoryProfile
    acceleration: float
    speed: Optional[float] = None
    duration_seconds: Optional[float] = None
```

Trajectory generation request parameters.

Parameter	Type	Description
<code>space</code>	<code>TrajectorySpace</code>	<code>JOINT</code> (joint space) or <code>CARTESIAN</code> (Cartesian space)
<code>frequency_hz</code>	<code>float</code>	Sampling frequency (Hz)

Parameter	Type	Description
<code>profile</code>	<code>TrajectoryProfile</code>	<code>CUBIC</code> (cubic polynomial) or <code>TRAPEZOIDAL</code>
<code>acceleration</code>	<code>float</code>	Acceleration
<code>speed</code>	<code>float</code>	Speed (mutually exclusive with <code>duration_seconds</code>)
<code>duration_seconds</code>	<code>float</code>	Total duration (mutually exclusive with <code>speed</code>)

Note: Exactly one of `speed` and `duration_seconds` must be set.

TrajectorySpace

```
class TrajectorySpace(Enum):
    JOINT = "Joint"
    CARTESIAN = "Cartesian"
```

TrajectoryProfile

```
class TrajectoryProfile(Enum):
    CUBIC = "Cubic"
    TRAPEZOIDAL = "Trapezoidal"
```

TrajectoryPoint

```
@dataclass
class TrajectoryPoint:
    t: float # Time (seconds)
    position: List[float] # 6 position values
```

Complete CRI Control Flow Example

```
from codroid import (
    CodroidClient,
    CriRealtimeDispatcher,
    TrajectoryGenerator,
    TrajectoryRequest,
    TrajectorySpace,
    TrajectoryProfile,
    InitConsoleUtf8,
)

InitConsoleUtf8()
```

```

ROBOT_IP = "192.168.8.136"

with CodroidClient(host=ROBOT_IP) as robot:
    robot.ConnectRemoteAndSwitchOn()

    # 1. Start CRI real-time control
    robot.StartCriControl()

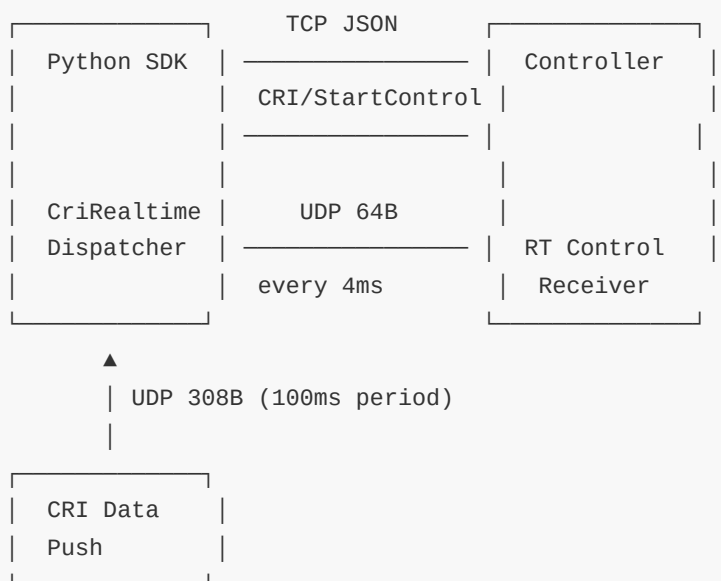
    # 2. Generate trajectory
    request = TrajectoryRequest(
        space=TrajectorySpace.JOINT,
        frequency_hz=250,
        profile=TrajectoryProfile.TRAPEZOIDAL,
        acceleration=100,
        speed=40,
    )
    trajectory = TrajectoryGenerator.generate(
        [0, 0, 0, 0, 0, 0],
        [0, 0, 90, 0, 90, 0],
        request,
    )

    # 3. Send trajectory
    with CriRealtimeDispatcher(ROBOT_IP) as dispatcher:
        dispatcher.SendTrajectory(trajectory, TrajectorySpace.JOINT, period_ms=4)

    # 4. Stop real-time control
    robot.StopCriControl()

```

Workflow Diagram



IO & Registers API Reference

IO Operations

Digital IO

GetDi

```
def GetDi(self, port: int) -> int
```

Get Digital Input (DI) value. Port 0~15, returns 0 or 1.

GetDo

```
def GetDo(self, port: int) -> int
```

Get Digital Output (DO) value. Port 0~15, returns 0 or 1.

SetDo

```
def SetDo(self, port: int, value: int) -> CommonResponse
```

Set Digital Output (DO) value. Port 0~15, value 0 or 1.

```
di0 = robot.GetDi(0)           # Read DI port 0
robot.SetDo(10, 1)             # Set DO port 10 to 1
robot.SetDo(10, di0)           # Write DI value to DO
```

Analog IO

GetAi

```
def GetAi(self, port: int) -> float
```

Get Analog Input (AI) value. Port 0~3.

GetAo

```
def GetAo(self, port: int) -> float
```

Get Analog Output (AO) value. Port 0~3.

SetAo

```
def SetAo(self, port: int, value: float) -> CommonResponse
```

Set Analog Output (AO) value. Port 0~3.

```
ai0 = robot.GetAi(0)           # Read AI port 0
robot.SetAo(0, 3.14)           # Set AO port 0
```

Batch IO

GetIoValues

```
def GetIoValues(self, io_requests: List[Dict[str, Any]]) -> CommonResponse
```

Batch read multiple IO values.

Parameter	Type	Description
<code>io_requests</code>	<code>List[Dict]</code>	List containing <code>type</code> and <code>port</code>

```
response = robot.GetIoValues([
    {"type": "DI", "port": 0},
    {"type": "DI", "port": 1},
    {"type": "AI", "port": 0},
])
for item in response.db:
    print(f"{item['type']}{item['port']} = {item['value']}")
```

SetIoValues

```
def SetIoValues(self, io_list: List[Dict[str, Any]]) -> List[CommonResponse]
```

Batch set IO values. Internally loops through `SetDo` / `SetAo`.

```
robot.SetIoValues([
    {"type": "DO", "port": 0, "value": 1},
    {"type": "DO", "port": 1, "value": 0},
    {"type": "AO", "port": 0, "value": 3.14},
])
```

IO Port Ranges

Type	Port Range	Value Type
DI (Digital Input)	0~15	0 or 1
DO (Digital Output)	0~15	0 or 1
AI (Analog Input)	0~3	float
AO (Analog Output)	0~3	float

Out-of-range values throw `CodroidError`.

Register Operations

GetRegisterValue

```
def GetRegisterValue(self, address: int) -> Any
```

Get a single register value.

GetRegisterValues

```
def GetRegisterValues(self, addresses: List[int]) -> CommonResponse
```

Batch get multiple register values.

```
val = robot.GetRegisterValue(0)
vals = robot.GetRegisterValues([0, 1, 2])
for item in vals.db:
    print(f"R{item['address']} = {item['value']}")
```

SetRegisterValue

```
def SetRegisterValue(self, address: int, value: Any) -> CommonResponse
```

Write register value.

```
robot.SetRegisterValue(0, 42)
robot.SetRegisterValue(1, 3.14)
```

Extended Arrays

SetExtendArrayType

```
def SetExtendArrayType(self, index: int, data_type: ExtendArrayType) -> CommonResponse
```

Set extended array data type. `index` range 0~999.

Data Type	Description
<code>ExtendArrayType.BOOL</code>	Boolean
<code>ExtendArrayType.UINT8</code>	Unsigned 8-bit integer
<code>ExtendArrayType.INT8</code>	Signed 8-bit integer
<code>ExtendArrayType.UINT16</code>	Unsigned 16-bit integer
<code>ExtendArrayType.INT16</code>	Signed 16-bit integer

Data Type	Description
<code>ExtendArrayType.UINT32</code>	Unsigned 32-bit integer
<code>ExtendArrayType.INT32</code>	Signed 32-bit integer
<code>ExtendArrayType.FLOAT32</code>	32-bit float

```
from codroid import ExtendArrayType

robot.SetExtendArrayType(0, ExtendArrayType.FLOAT32)
```

RemoveExtendArray

```
def RemoveExtendArray(self, index: int) -> CommonResponse
```

Remove extended array index (reset data). `index` range 0~999.

Complete IO + Register Example

```
from codroid import CodroidClient, ExtendArrayType, InitConsoleUtf8

InitConsoleUtf8()

with CodroidClient(host="192.168.8.136") as robot:
    robot.ConnectRemoteAndSwitchOn()

    # --- IO ---
    # Read single DI
    di0 = robot.GetDi(0)
    print(f"DI 0 = {di0}")

    # Write single DO
    robot.SetDo(10, 1)

    # Batch read
    io_data = robot.GetIoValues([
        {"type": "DI", "port": 0},
        {"type": "DI", "port": 1},
        {"type": "AI", "port": 0},
    ])
    for item in io_data.db:
        print(f" {item['type']}{item['port']} = {item['value']}")

    # --- Registers ---
    # Read single
    val = robot.GetRegisterValue(0)
    print(f"R0 = {val}")

    # Write
```

```
robot.SetRegisterValue(0, val + 1)

# Batch read
regs = robot.GetRegisterValues([0, 1, 2])
for item in regs.db:
    print(f" R{item['address']} = {item['value']}")

# Extended arrays
robot.SetExtendArrayType(0, ExtendArrayType.FLOAT32)
robot.RemoveExtendArray(0)
```

Utilities API Reference

Publish / Subscribe

PublishTopics

```
class PublishTopics:
    PROJECT_STATE = "publish/ProjectState"
    VAR_UPDATE = "publish/VarUpdate"
    ROBOT_STATUS = "publish/RobotStatus"
    ROBOT_POSTURE = "publish/RobotPosture"
    ROBOT_COORDINATE = "publish/RobotCoordinate"
    LOG = "publish/Log"
    ERROR = "publish/Error"
```

Controller push topic constants.

PublishNotification

```
@dataclass
class PublishNotification:
    ty: str          # Topic type
    db: Any          # Push data
    raw_json: str    # Raw JSON string
```

PublishTopicSubscription

```
class PublishTopicSubscription:
    def dispose(self) -> None
    def Dispose(self) -> None # C# alias
```

Subscription handle. Call `dispose()` to remove local callback (does not send unsubscribe to controller).

SubscribePublishTopic (CodroidClient only)

```
def SubscribePublishTopic(
    self,
    topic_ty: str,
    handler: Callable[[PublishNotification], None],
    tc_milliseconds: int = 100,
) -> PublishTopicSubscription
```

Subscribe to controller push topic. On first local subscription, automatically sends subscription request to controller.

```
from codroid import CodroidClient, PublishTopics, InitConsoleUtf8
```

```

InitConsoleUtf8()

def on_status(notification):
    print(f"Received {notification.ty}: {notification.db}")

def on_error(notification):
    print(f"Error: {notification.db}")

with CodroidClient(host="192.168.8.136") as robot:
    robot.ConnectRemoteAndSwitchOn()

    sub1 = robot.SubscribePublishTopic(PublishTopics.ROBOT_STATUS, on_status)
    sub2 = robot.SubscribePublishTopic(PublishTopics.ERROR, on_error)

    # ... run ...

    sub1.dispose()
    sub2.dispose()

```

Global Variables

GetGlobalVars

```
def GetGlobalVars(self) -> CommonResponse
```

Get all global variables.

GetGlobalVarsCatalog

```
def GetGlobalVarsCatalog(self) -> CommonResponse
```

Get global variables catalog (same TCP request as `GetGlobalVars`).

SaveGlobalVar

```
def SaveGlobalVar(self, name: str, variable: GlobalVariable) -> CommonResponse
```

Save a single global variable.

SaveGlobalVars

```
def SaveGlobalVars(self, variables: Dict[str, GlobalVariable]) -> CommonResponse
```

Batch save global variables. Variable names are validated against Lua reserved words.

RemoveGlobalVars

```
def RemoveGlobalVars(self, names: List[str]) -> CommonResponse
```

Remove global variables.

GetProjectVar

```
def GetProjectVar(self) -> CommonResponse
```

Get all current project variable values (only valid when project is running).

```
from codroid import GlobalVariable

# Save
robot.SaveGlobalVar("counter", GlobalVariable(value=0, note="counter"))
robot.SaveGlobalVar("name", GlobalVariable(value="test"))

# Batch save
robot.SaveGlobalVars({
    "x": GlobalVariable(value=100.0),
    "y": GlobalVariable(value=200.0),
})

# Read
response = robot.GetGlobalVars()
print(response.db)

# Remove
robot.RemoveGlobalVars(["counter", "name"])

# Get project variables (while project is running)
proj_vars = robot.GetProjectVar()
```

Kinematics

AposToCpos

```
def AposToCpos(
    self,
    jp: Sequence[float],
    coor: Optional[Sequence[float]] = None,
    tool: Optional[Sequence[float]] = None,
    ep: Sequence[float] = [],
) -> CommonResponse
```

Forward kinematics (joint → Cartesian). `jp` is 6 joint angles (degrees).

Parameter	Type	Description
<code>jp</code>	<code>Sequence[float]</code>	6 joint angles (degrees)
<code>coor</code>	<code>Sequence[float]</code>	User coordinate <code>[x, y, z, a, b, c]</code>
<code>tool</code>	<code>Sequence[float]</code>	Tool coordinate <code>[x, y, z, a, b, c]</code>
<code>ep</code>	<code>Sequence[float]</code>	External axis positions

```
response = robot.AposToCpos([0, 0, 90, 0, 90, 0])
print(f"TCP pose: {response.db}")
```

CposToApos

```
def CposToApos(
    self,
    cp: Sequence[float],
    rj: Optional[Sequence[float]] = None,
    ep: Sequence[float] = [],
) -> CommonResponse
```

Inverse kinematics (Cartesian → joint). `cp` is `[x, y, z, a, b, c]` (mm + degrees). `rj` is reference joint angles (default `[20, 20, 20, 20, 20, 20]`).

```
response = robot.CposToApos([400, 200, 500, 180, 0, 90])
print(f"Joint angles: {response.db}")
```

CalculateRelativePose

```
def CalculateRelativePose(
    self,
    pos: Sequence[float],
    offset: Sequence[float],
    coor_type: CoordinateType = CoordinateType.TOOL,
    pos_coor: Optional[Sequence[float]] = None,
    coor: Optional[Sequence[float]] = None,
) -> CommonResponse
```

Cartesian coordinate offset calculation.

Parameter	Type	Description
<code>pos</code>	<code>Sequence[float]</code>	Current TCP pose
<code>offset</code>	<code>Sequence[float]</code>	Offset values
<code>coor_type</code>	<code>CoordinateType</code>	<code>USER</code> or <code>TOOL</code>
<code>pos_coor</code>	<code>Sequence[float]</code>	Current TCP coordinate

Parameter	Type	Description
<code>coord</code>	<code>Sequence[float]</code>	Offset coordinate

```
from codroid import CoordinateType

response = robot.CalculateRelativePose(
    pos=[400, 200, 500, 180, 0, 90],
    offset=[0, 0, -100, 0, 0, 0],
    coord_type=CoordinateType.TOOL,
)
print(f"Offset pose: {response.db}")
```

CposToCpos / CposToCposPose (2.1.7+)

Coordinate system transformation: convert a TCP pose from coordinate system 1 + tool 1 to coordinate system 2 + tool 2. Protocol `Robot/cpostocpos`.

```
def CposToCpos(
    self,
    cp: Sequence[float],
    coord1: Sequence[float],
    tool1: Sequence[float],
    coord2: Sequence[float],
    tool2: Sequence[float],
) -> List[float]
```

Parameter	Type	Description
<code>cp</code>	<code>Sequence[float]</code>	Current TCP pose <code>[x, y, z, a, b, c]</code> (mm+deg)
<code>coord1</code>	<code>Sequence[float]</code>	Source coordinate system <code>[x, y, z, a, b, c]</code>
<code>tool1</code>	<code>Sequence[float]</code>	Source tool <code>[x, y, z, a, b, c]</code>
<code>coord2</code>	<code>Sequence[float]</code>	Target coordinate system <code>[x, y, z, a, b, c]</code>
<code>tool2</code>	<code>Sequence[float]</code>	Target tool <code>[x, y, z, a, b, c]</code>

```
result = robot.CposToCpos(
    cp=[400, 200, 500, 180, 0, 90],
    coord1=[0,0,0,0,0,0], tool1=[0,0,0,0,0,0],
    coord2=[100,0,0,0,0,0], tool2=[0,0,100,0,0,0]
)
print(f"Converted pose: {result}")
```

`CposToCposPose` returns `CartesianPoint`:

```
pose = robot.CposToCposPose(
    cp=CartesianPoint.MmDeg([400, 200, 500, 180, 0, 90]),
    coor1=[0,0,0,0,0,0], tool1=[0,0,0,0,0,0],
    coor2=[100,0,0,0,0,0], tool2=[0,0,100,0,0,0]
)
```

ConsoleUtf8

InitConsoleUtf8

```
def InitConsoleUtf8() -> None
```

Windows console UTF-8 initialization. Must be called at entry when running examples with Chinese characters in `cmd` (not Windows Terminal). No-op on Linux / macOS.

```
from codroid import InitConsoleUtf8

InitConsoleUtf8()
```

Legacy alias: `init_console_utf8 = InitConsoleUtf8`

PrintBanner

```
from codroid import PrintBanner

PrintBanner("Title", "Subtitle")
```

Print a colored banner title in the terminal. Requires `colorama`: `pip install "codroid-robot-sdk[color]"`.

Complete Utilities Example

```
from codroid import (
    CodroidClient,
    PublishTopics,
    GlobalVariable,
    CoordinateType,
    InitConsoleUtf8,
    PrintBanner,
)

InitConsoleUtf8()
PrintBanner("Utilities Example", "Publish / Global Vars / Kinematics")

ROBOT_IP = "192.168.8.136"

# --- Publish / Subscribe ---
```

```

def on_status(notification):
    print(f"[Publish] {notification.ty}: {notification.db}")

with CodroidClient(host=ROBOT_IP) as robot:
    robot.ConnectRemoteAndSwitchOn()

    sub = robot.SubscribePublishTopic(PublishTopics.ROBOT_STATUS, on_status)

    # --- Global Variables ---
    robot.SaveGlobalVar("test_var", GlobalVariable(value=42, note="test"))
    response = robot.GetGlobalVars()
    print(f"Global vars: {response.db}")
    robot.RemoveGlobalVars(["test_var"])

    # --- Kinematics ---
    fk = robot.AposToCpos([0, 0, 90, 0, 90, 0])
    print(f"FK result: {fk.db}")

    ik = robot.CposToApos([400, 200, 500, 180, 0, 90])
    print(f"IK result: {ik.db}")

    rel = robot.CalculateRelativePose(
        pos=[400, 200, 500, 180, 0, 90],
        offset=[0, 0, -100, 0, 0, 0],
        coord_type=CoordinateType.T00L,
    )
    print(f"Offset result: {rel.db}")

    sub.dispose()

```